

Problem 1

```
x = [1, 2, 3, 4, 5, 6, 7, 8]
```

```
x = 1×8  
    1    2    3    4    5    6    7    8
```

```
x_fft = fft(x)
```

```
x_fft = 1×8 complex  
    36.0000 + 0.0000i   -4.0000 + 9.6569i   -4.0000 + 4.0000i   -4.0000 + 1.6569i   -4.0000 + 0.0000i   -4.0000 + 0.0000i   -4.0000 + 0.0000i   -4.0000 + 0.0000i
```

```
x_ifft = ifft(x_fft)
```

```
x_ifft = 1×8  
    1    2    3    4    5    6    7    8
```

```
x_myfft = myFFT(x)
```

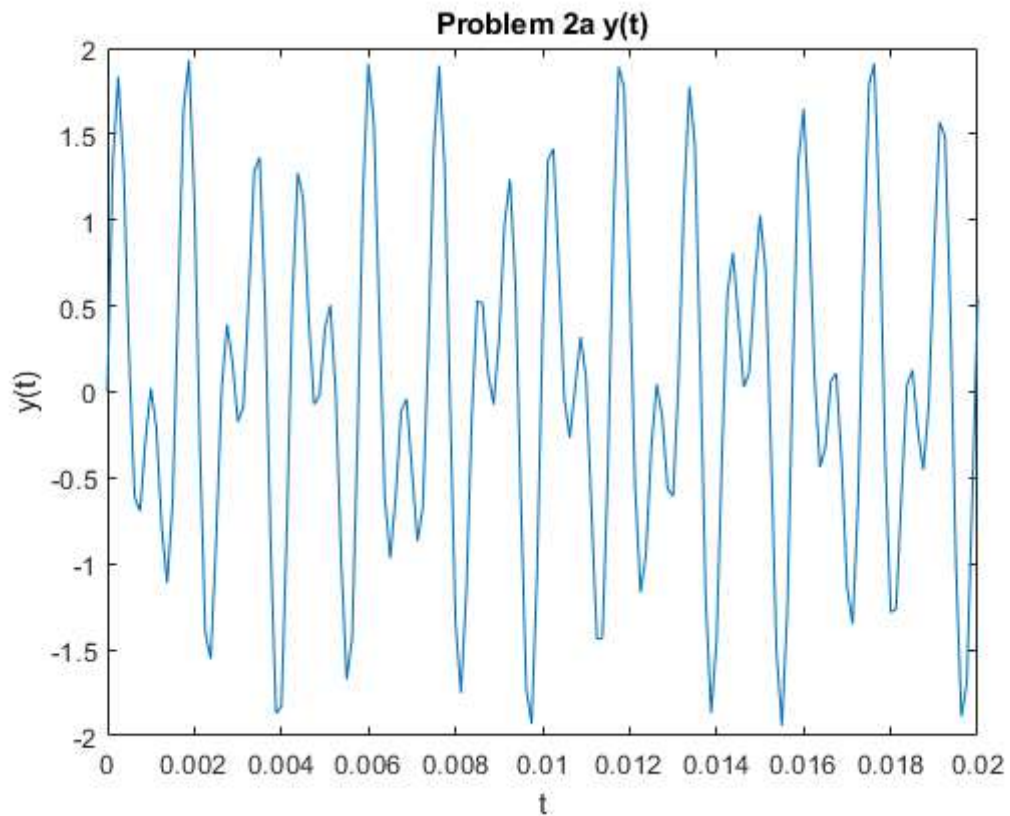
```
x_myfft = 1×8 complex  
    36.0000 + 0.0000i   -4.0000 + 9.6569i   -4.0000 + 4.0000i   -4.0000 + 1.6569i   -4.0000 - 0.0000i   -4.0000 - 0.0000i   -4.0000 - 0.0000i   -4.0000 - 0.0000i
```

```
x_myifft = myIFFT(x_myfft)
```

```
x_myifft = 1×8 complex  
    1.0000 - 0.0000i    2.0000 - 0.0000i    3.0000 - 0.0000i    4.0000 - 0.0000i    5.0000 - 0.0000i    6.0000 - 0.0000i    7.0000 - 0.0000i    8.0000 - 0.0000i
```

Problem 2

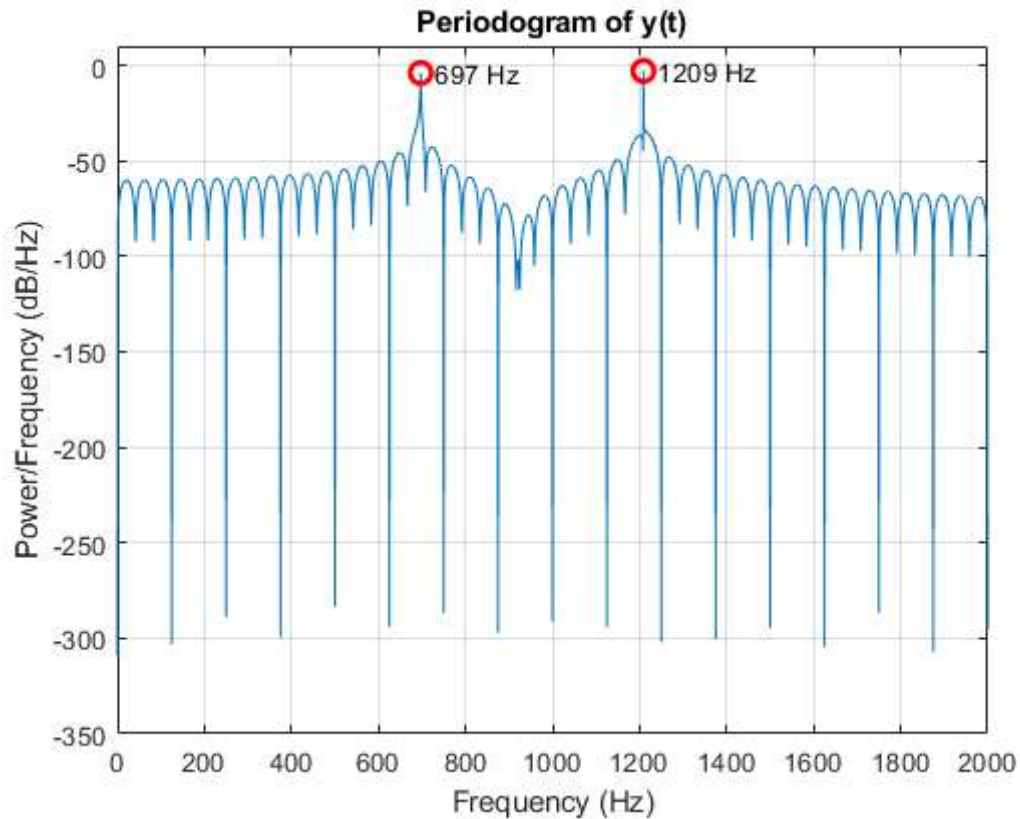
```
fs = 8000; % sample rate in Hz  
dt = 1/fs; % sampling interval  
t = 0:dt:1;  
  
f1 = 697; % Hz  
f2 = 1209; % Hz  
  
y = sin(2*pi*f1*t) + sin(2*pi*f2*t);  
plot(t, y)  
axis([0 0.02 -2 2])  
title('Problem 2a y(t)');  
xlabel('t');  
ylabel('y(t)');
```



```
[pxx, f] = periodogram(y, [], [], fs);
pxx_db = 10*log10(pxx);

plot(f, pxx_db)
title('Periodogram of y(t)')
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
xlim([0 2000]);
ylim([-350 10])
grid on
hold on

[~, i1] = min(abs(f - 697));
[~, i2] = min(abs(f - 1209));
plot(f(i1), pxx_db(i1), 'ro', 'MarkerSize', 8, 'LineWidth', 2)
plot(f(i2), pxx_db(i2), 'ro', 'MarkerSize', 8, 'LineWidth', 2)
text(f(i1), pxx_db(i1), ' 697 Hz')
text(f(i2), pxx_db(i2), ' 1209 Hz')
hold off
```



```
% signalAnalyzer(y)
% screenshot of signal analyzer at the bottom of the pdf

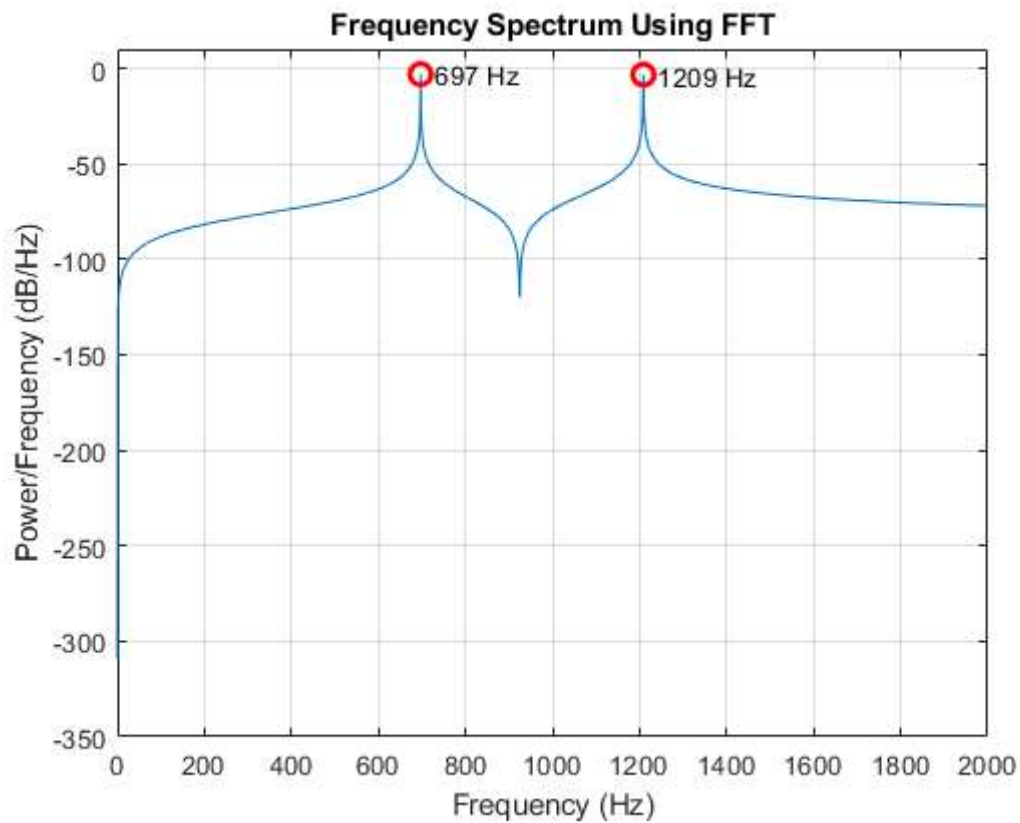
N = length(y);
Y = fft(y);

P2 = abs(Y/N).^2;
P1 = P2(1:floor(N/2)+1);
P1(2:end-1) = 2*P1(2:end-1);
P1_dB = 10*log10(P1);
f = fs*(0:floor(N/2))/N;

plot(f, P1_dB)
xlabel('Frequency (Hz)')
ylabel('Power/Frequency (dB/Hz)')
title('Frequency Spectrum Using FFT')
axis([0 2000 -350 10])
grid on
hold on

[~, i1] = min(abs(f - f1));
[~, i2] = min(abs(f - f2));
plot(f(i1), P1_dB(i1), 'ro', 'MarkerSize', 8, 'LineWidth', 2)
plot(f(i2), P1_dB(i2), 'ro', 'MarkerSize', 8, 'LineWidth', 2)
text(f(i1), P1_dB(i1), ' 697 Hz')
text(f(i2), P1_dB(i2), ' 1209 Hz')

hold off
```



Problem 1 Defined FFT and IFFT Functions

```
function X = myFFT(x)
    N = length(x);
    W = exp(-1j*(2*pi/N));

    X = zeros(1,N);

    for k = 0:N-1
        for n = 0:N-1
            X(k+1) = X(k+1) + x(n+1)*W^(k*n);
        end
    end
end

function x = myIFFT(X)
    N = length(X);
    W = exp(-1j*(2*pi/N));

    x = zeros(1,N);

    for k = 0:N-1
        for n = 0:N-1
            x(n+1) = x(n+1) + X(k+1)*W^(-1*k*n);
        end
    end

    x = x/N;
```

end